Progress Report

Discrete State-Space Models and Expectation-Maximization for Cell-State Trajectories

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Overview

In weeks 7–8 I implemented the population model and its EM algorithm, and tested them on simulated cells from the 2D toy model of the last report.

- The code matches the hand-check values, and ℓ never decreases during EM.
- On simulated cells the MA and MU rows of A are recovered well, the T row worst.
- EM often stops at a lower ℓ (a local maximum), so I fit from 21 starts and keep the fit with the highest ℓ.

Done from the plan of the last report:

1. Implement the 2D toy population model and pass the hand-check values (Section 1).
2. Run the recovery experiment on the 2D toy (Sections 2–3).

The rest of that plan (the day-0 experiment, the 5D toy and the first look at the real data) is in the plan for the next two weeks (Section 4).

1 Implementation and checks

The model and EM are in `population_model.py`, the starting values in `starting_values.py`, and the true toy parameters in `toy_parameters.py`. The code is in the GitHub repository https://github.com/bdtiger/open-course-on-topics (folder `project-code`). The fitting code gets only the expression matrix X (one row per cell, one column per gene) and the week of each cell. For simulated cells, the true state of each cell is kept aside and used only to compare afterwards. So the same code can run on the real data.

One fix was needed: in the π, A step a probability could round to exactly 0 in floating point, and after that π and A stopped changing. Every entry of π and A is now kept at 10^{-10} or above.

The checks in `checks.py` all pass (Table 1). Two of them cannot expect exact equality, so they pass if the gap stays below a limit I chose. For the simulator, random sampling never matches the parameters exactly; the limits are 0.02 for the state fractions and 0.1 for the means and variances, a few times the random error expected with 3000 cells per week and much smaller than the error a bug would cause. For the π, A step, the numerical optimiser stops close to the optimum but not exactly on it; the limit is 10^{-3}. The table gives the actual gaps.

Check	Compared with	Result
Proportions p_1, p_2, p_3	hand values	equal
Simulator, 3000 cells per week	p_t, μ_j, σ_j^2	largest gap: fractions 0.009, means and variances 0.02
$\log \mathcal{N}$, r and ℓ of the four hand-check cells	hand values	equal to 4 decimals
One Gaussian update from the hand-check cells	hand values	equal to 4 decimals
π, A step with exact proportions $n_t = 10^5\, p_t$	true π, A	largest gap 2×10^{-7}
State matching on the 6 reorderings of the true parameters	true order	order restored exactly
EM (every iteration)	previous ℓ	ℓ never decreases

Table 1: Checks of the implementation. Hand values are from the hand-check sheet of the last report.

2 Several starts

What I found

EM only climbs: every iteration raises ℓ, and it stops when ℓ no longer rises. Where it stops depends on where it starts, and from many starts it stops at a lower ℓ (a local maximum). For example, two states can share the MA cells while no state covers T.

I therefore fit from 21 starts and keep the fit with the highest final ℓ:

- the Householder start from the last report (k-means means and variances, A from the Householder formula);
- 20 random starts (random π and A, three random cells as the means, the variance of all cells as the variances).

Figure 1 shows ℓ for every start. Only 2 of the 21 starts at 300 cells per week, and 5 of 21 at 3000, reach the best ℓ (within 0.1). The Householder start does not reach it at either size. At 3000 cells per week the first random start that reaches it is number 10, so the five random starts I had planned would have missed it. Some starts stay flat for hundreds of iterations before they climb again (blue line, left), so I allow up to 2000 iterations per start; one start at 300 cells per week was still climbing slowly when it reached this limit.

In the project

On the real data there is no true answer to compare with, so fits can only be compared by ℓ. I will run many random starts, keep the best ℓ, and report how many starts reach it.

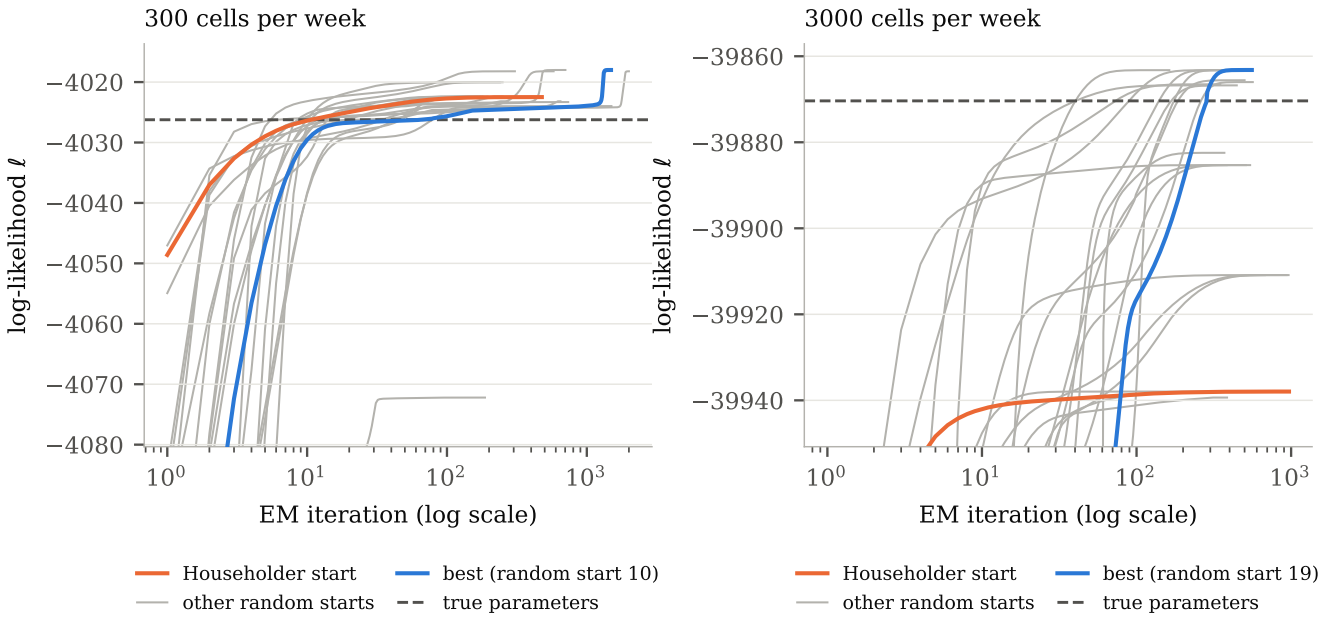


Figure 1: ℓ per EM iteration for every start (iteration axis logarithmic). Blue: the start with the highest final ℓ. Orange: the Householder start. Gray: the other random starts; each flat end is a local maximum. Dashed: ℓ of the true parameters.

3 Recovery of the true parameters

What I did

I simulated 300 and 3000 cells per week for the four weeks from the true toy parameters (Figure 2), fitted them as in Section 2, and compared the best fit with the true parameters. EM numbers its states in no particular order, so before comparing I name each fitted state after the true state with the closest mean.

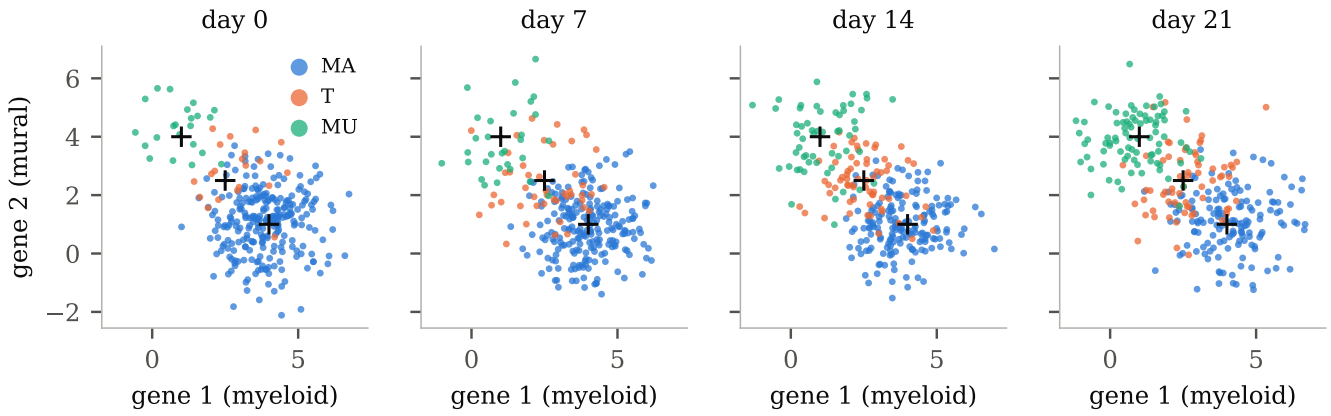


Figure 2: Simulated cells, 300 per week, coloured by their true state (the fit never sees the states). Crosses: true means. Over the weeks the population moves from MA towards T and MU.

What I found

Figure 3 shows the results.

- The MA and MU rows of A are recovered well, the T row worst. The largest error $|\text{fitted} - \text{true}|$ in each row is 0.05 (MA), 0.21 (T) and 0.05 (MU) at 300 cells per week, and 0.04, 0.13 and 0.05 at 3000. More cells improve the MA and T rows; the MU row has the same error at both sizes.
- The last report (Section 6) predicted the T and MU rows to be less accurate than the MA row. This holds for T but not for MU.
- The best fit has a slightly higher ℓ than the true parameters at both sizes (-4018.00 against -4026.23 at 300 cells per week, -39863.19 against -39870.38 at 3000). This is expected: the fit also adapts to the random noise in this particular sample of cells.

In the project

On the real data I expect the T row to be the least certain. Before I interpret it, I will report how much each row of A changes across starts and resampled cells, as planned in the last report.

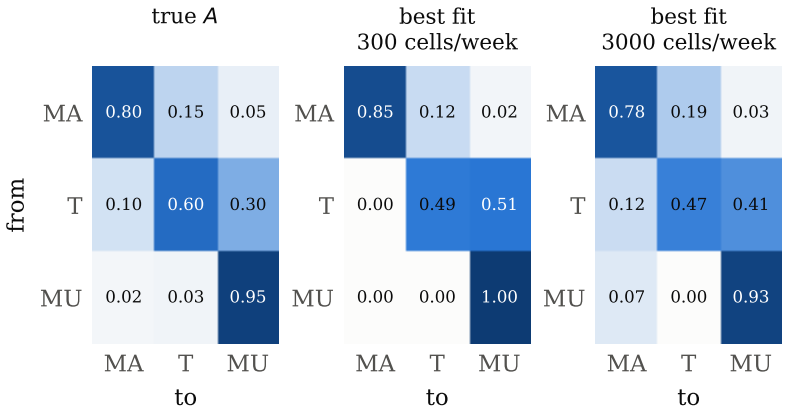


Figure 3: True A and the A of the best fit at 300 and 3000 cells per week. Rows: from; columns: to. Each fitted state is named after the true state with the closest mean.

4 Plan for the next two weeks

1. Finish the toy experiments:
   - Day-0 experiment: refit the simulated cells without the day-0 cells, from several starts, to check that A cannot be estimated without the healthy cells (expected: different starts end at different A with almost the same ℓ).
   - Test set: keep 20% of the cells aside from fitting, and compare their ℓ under the best fit with their ℓ under the true parameters.
   - The 5D toy with one gene that carries no signal.
2. Real data, first look only: open `adata_final.h5ad`, select the 50 genes, check the normalisation, and count the cells per day, batch and cluster, before any fitting.